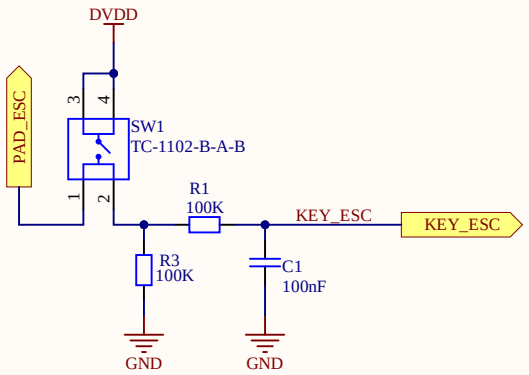
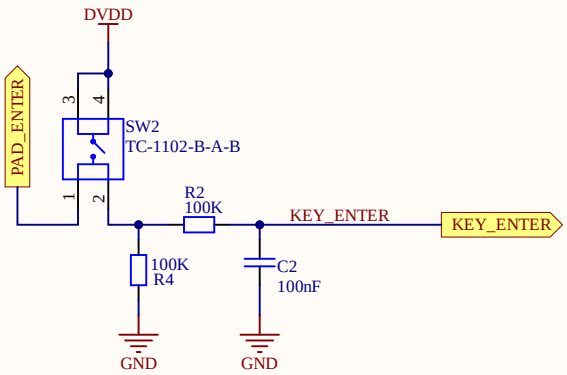
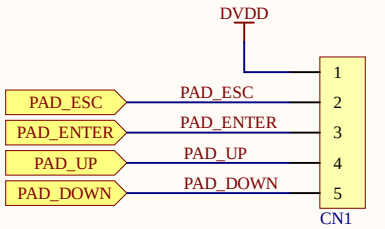
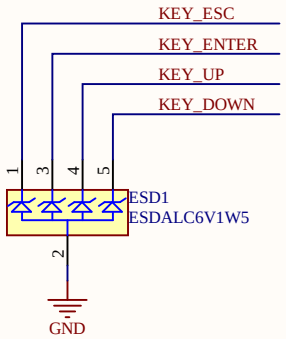
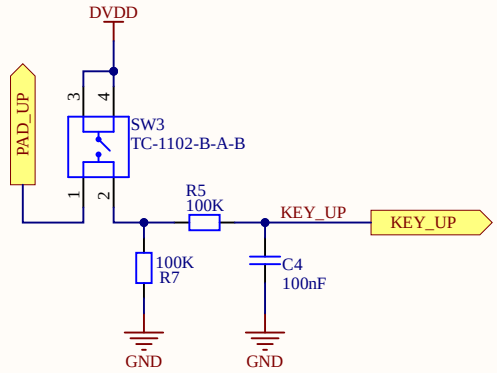
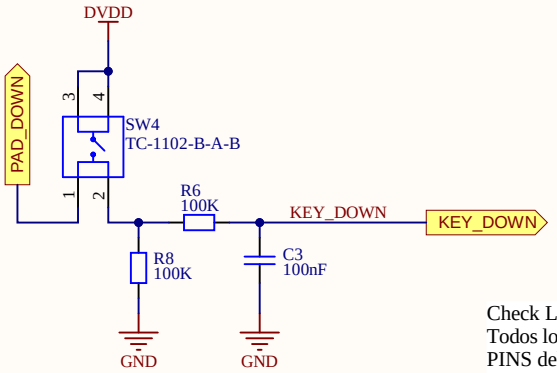








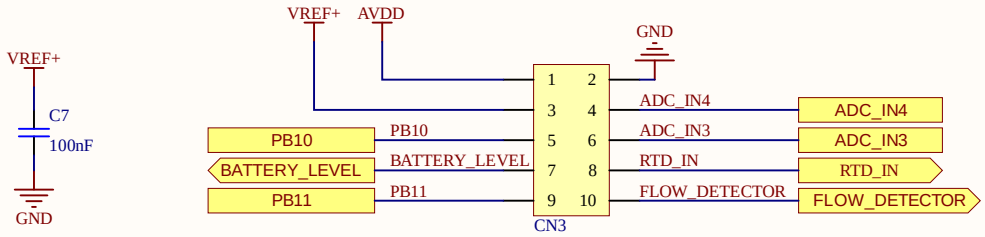


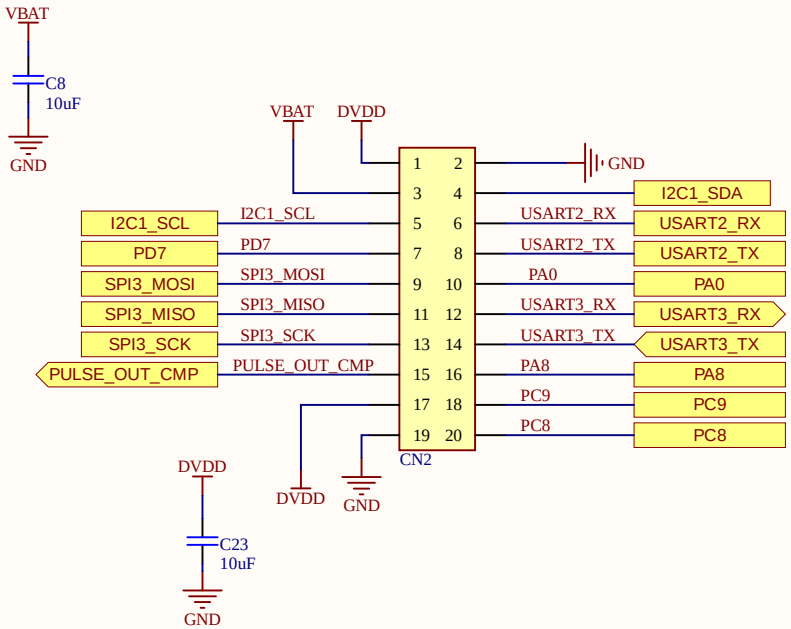


Todas las hojas de datos en la carpeta datasheet en la libreria de los componentes altium

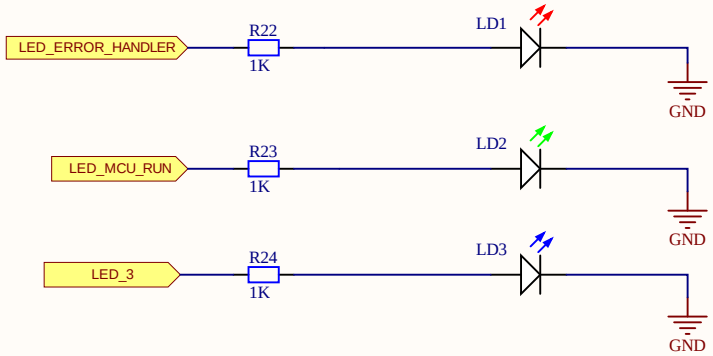
Todos los datasheet en la carpeta datasheet de la libreria altium

Check List:
Todos los componentes en milimetros, PINs en grilla de 2mm -> OK
PINS del MC expouestos con proteccion ESD -> OK
Filtro anti-rebotenes para switchs -> OK
Uso de simbols de alimentacion adecuados -> OK
Uso de y tierra adecuadas -> OK
Compatibilidad conector externo de teclado con versiones anteriores -> Pendiente
Firmare testeo de funcionamineto teclado -> Pendiente
Parametros basicos de los componentes cargados -> Pendiente





Los LEDs con un jumper en serie tienen proposito solo de debug



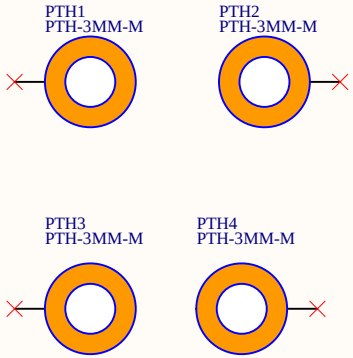


Tabla de contenidos

1 - Esta pagina

2- CON1, conectores placa fm-320-u-1

3 - Teclado, coneccion a keypad y led de señalizacion

4 - LCD, driver Ly back light

5 - STM32U575 alimentacion

6 - STM32U575 puertos

5 - STM32U575 debugger y programacion

Tabla de contenidos

Comentarios generales, titulos, nombres de bloques

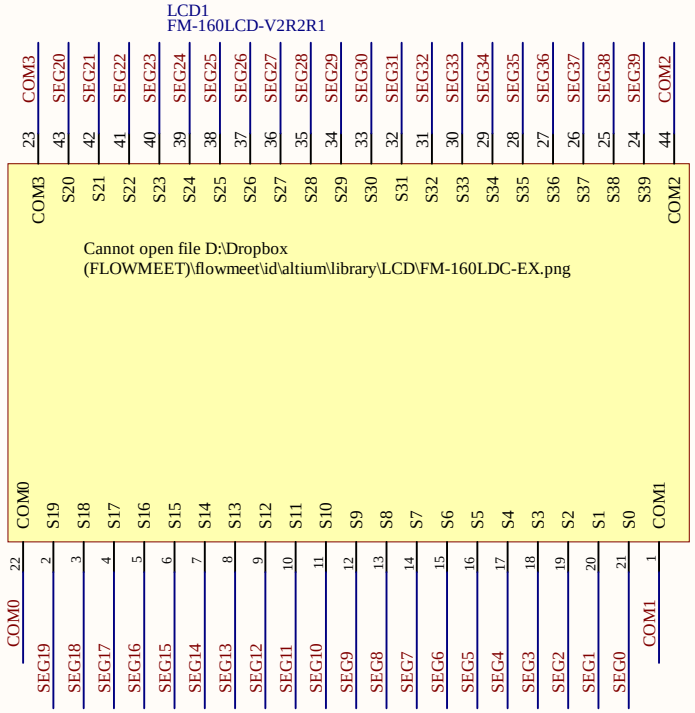
Texto para ser agregado en el silkscreen

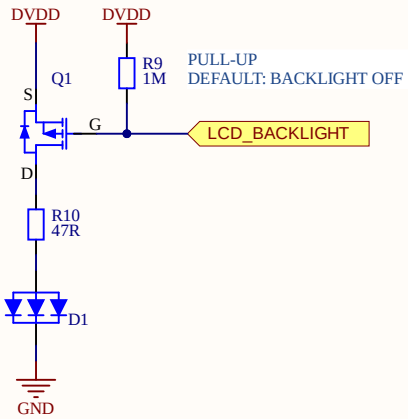
Advertencias

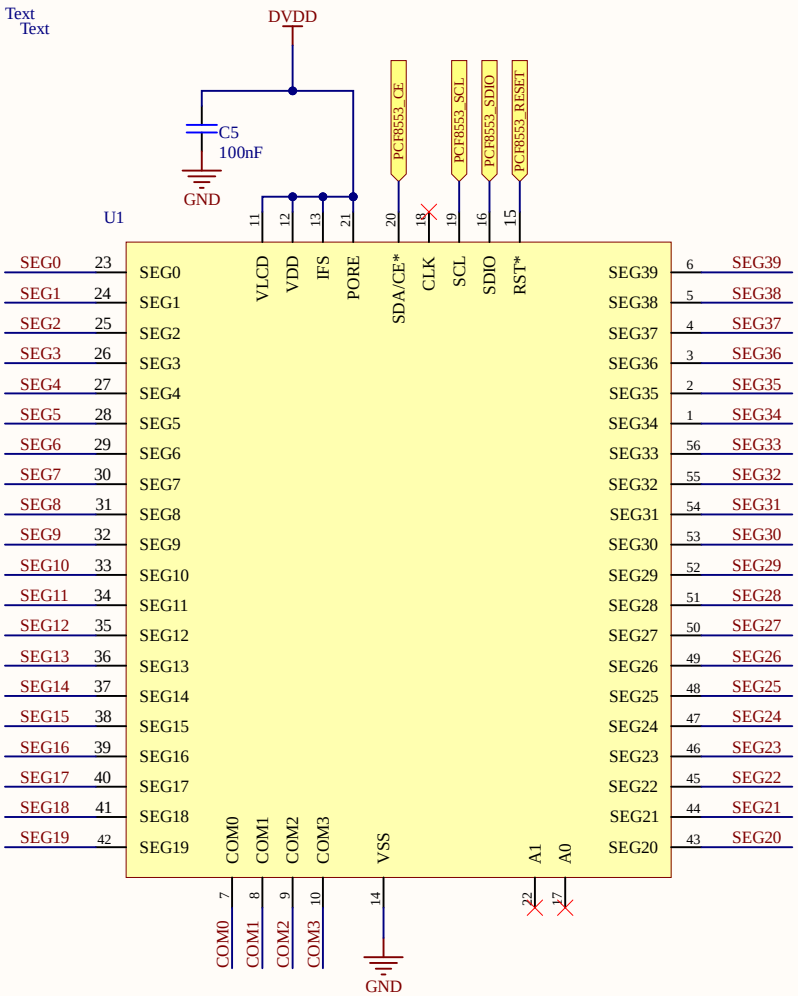
Directivas a tener en cuenta al genera el PCB

PENDIENTES ANTES DE FABRICAR EL PCB
- Actualizar operacional cuadruple en proyecto fmc-320u-b.
- Firmare testeo de funcionamineto teclado
- Capacitodes de des-acople suficientes, agregar en los headers

Title: FMC-320U - MODELO A - PCB 1		Version: 1	FLOWMEET SRL
Size: A4	Sheet: 1 Of: 7	Revision: 1	
Date: 9/11/2023	Time: 16:20:40	Release: 1	
File D:\FLOWMEET Dropbox\flowmeet\id\equipos\fmc-320u\hardware\altium\model_a\trunk\FMC-320U-A1\READ_1.SchDoc			



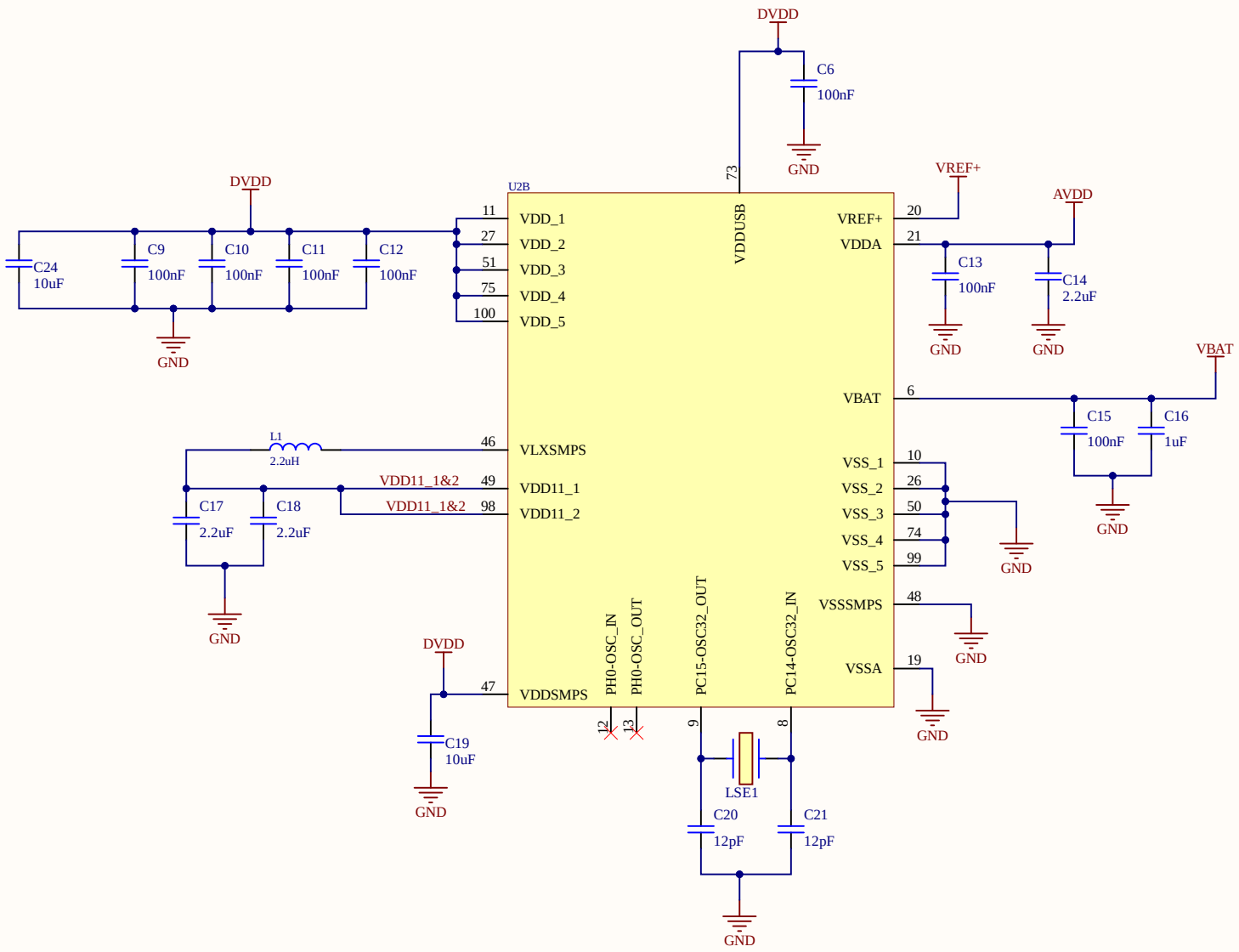




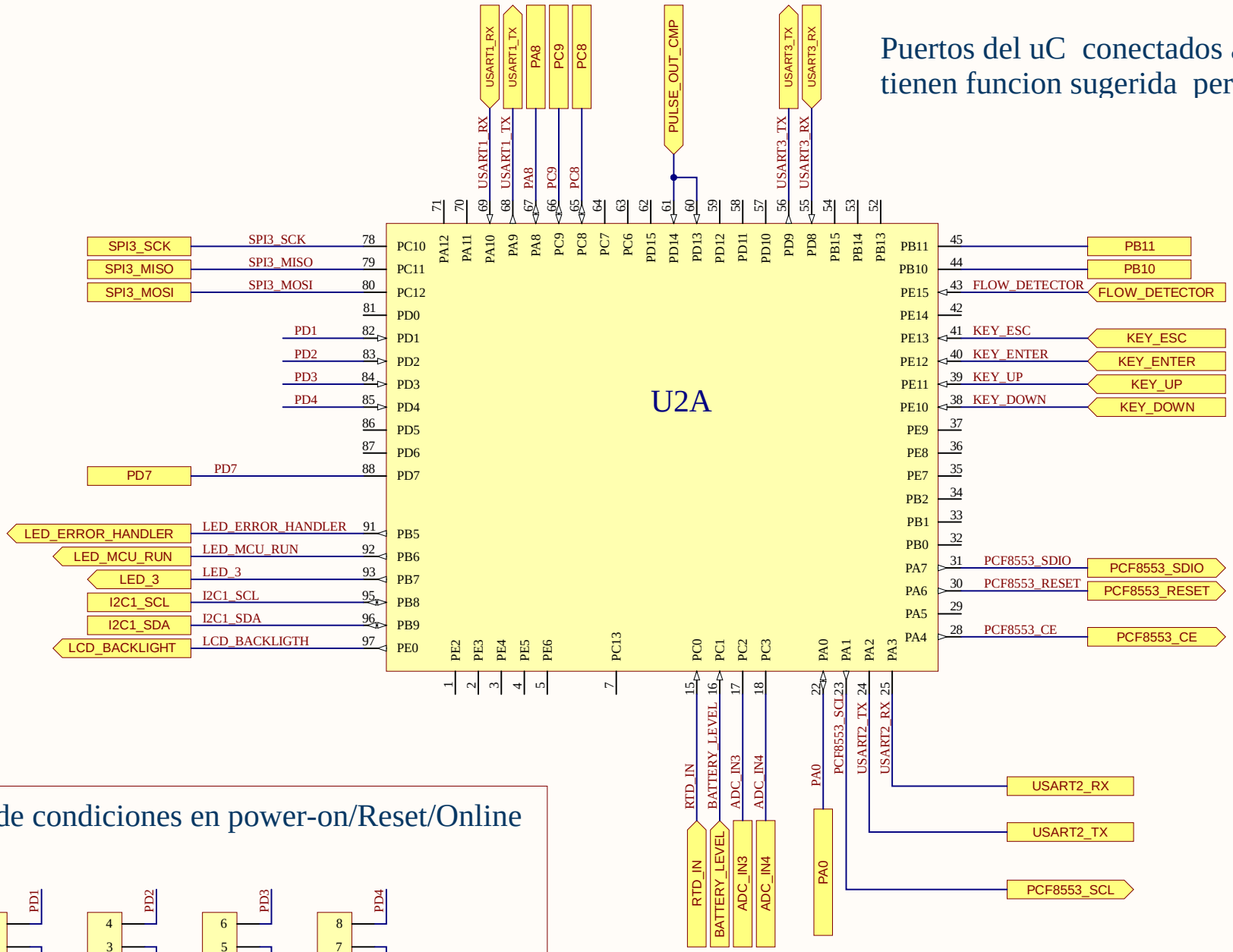
POTENCIA DISPADA

R10 = (DVDD -VLED)^2 / 47 = (3-2)^2 / 47 = 1/47

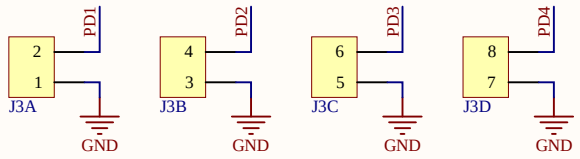
A0 y A1 tienen pull-down interno.

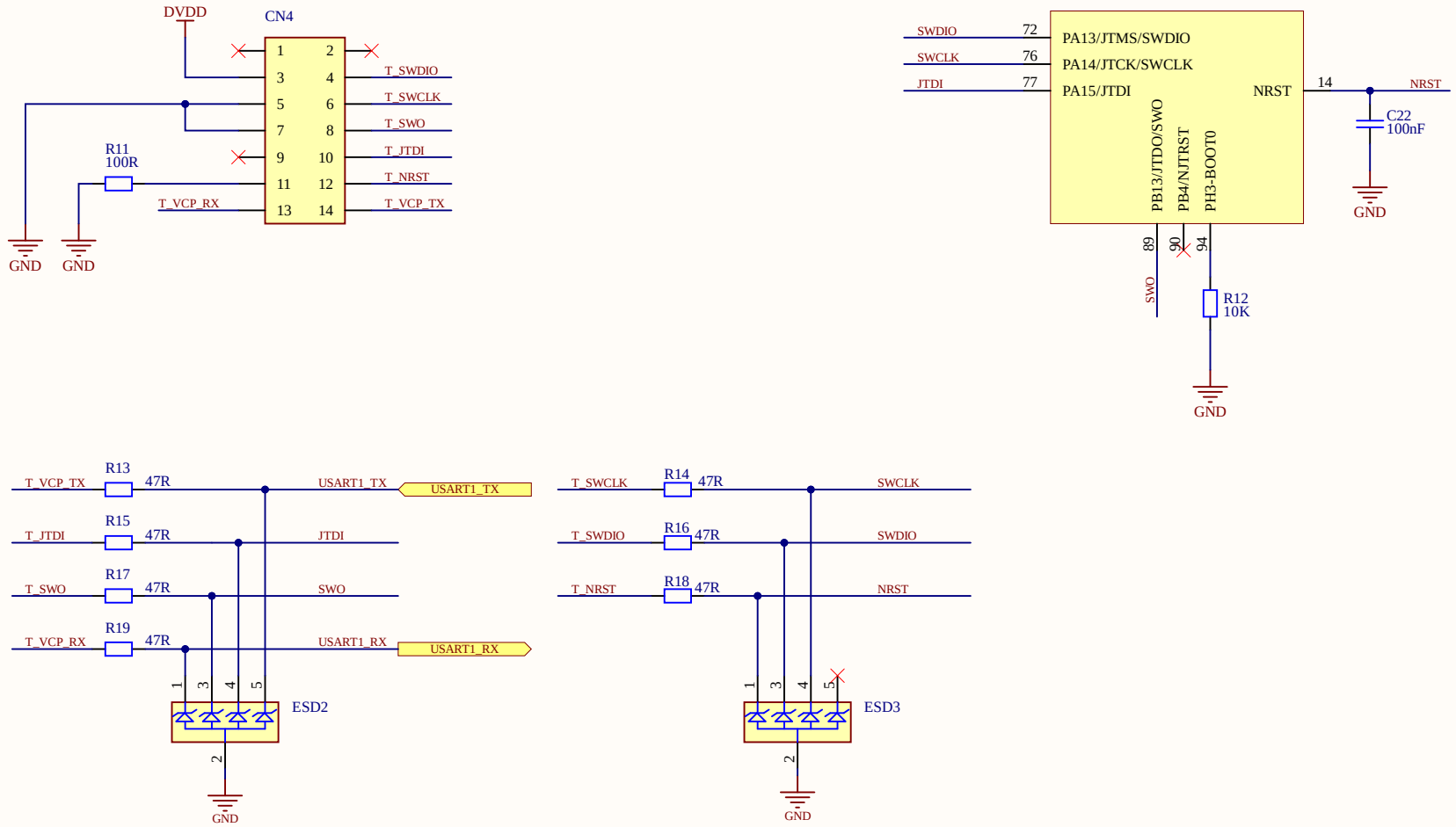

DVDD
C6
100nF
GND
73
VDDUSB
U2B
DVDD
C24
10uF
C9
100nF
C10
100nF
C11
100nF
C12
100nF
GND
11
27
51
75
100
VDD_1
VDD_2
VDD_3
VDD_4
VDD_5
VREF+
20
VREF+
21
VDDA
AVDD
C13
100nF
C14
2.2uF
GND
GND
VBAT
6
VBAT
C15
100nF
C16
1uF
GND
L1
2.2uH
46
VLXSMPS
VDD11_1&2
49
VDD11_1
VDD11_1&2
98
VDD11_2
C17
2.2uF
C18
2.2uF
GND
10
26
50
74
99
VSS_1
VSS_2
VSS_3
VSS_4
VSS_5
GND
48
VSSSMPS
GND
19
VSSA
GND
DVDD
47
VDDSMPS
C19
10uF
GND
PH0-OSC_IN
PH0-OSC_OUT
PC15-OSC32_OUT
PC14-OSC32_IN
12
13
9
8
LSE1
C20
12pF
C21
12pF
GND

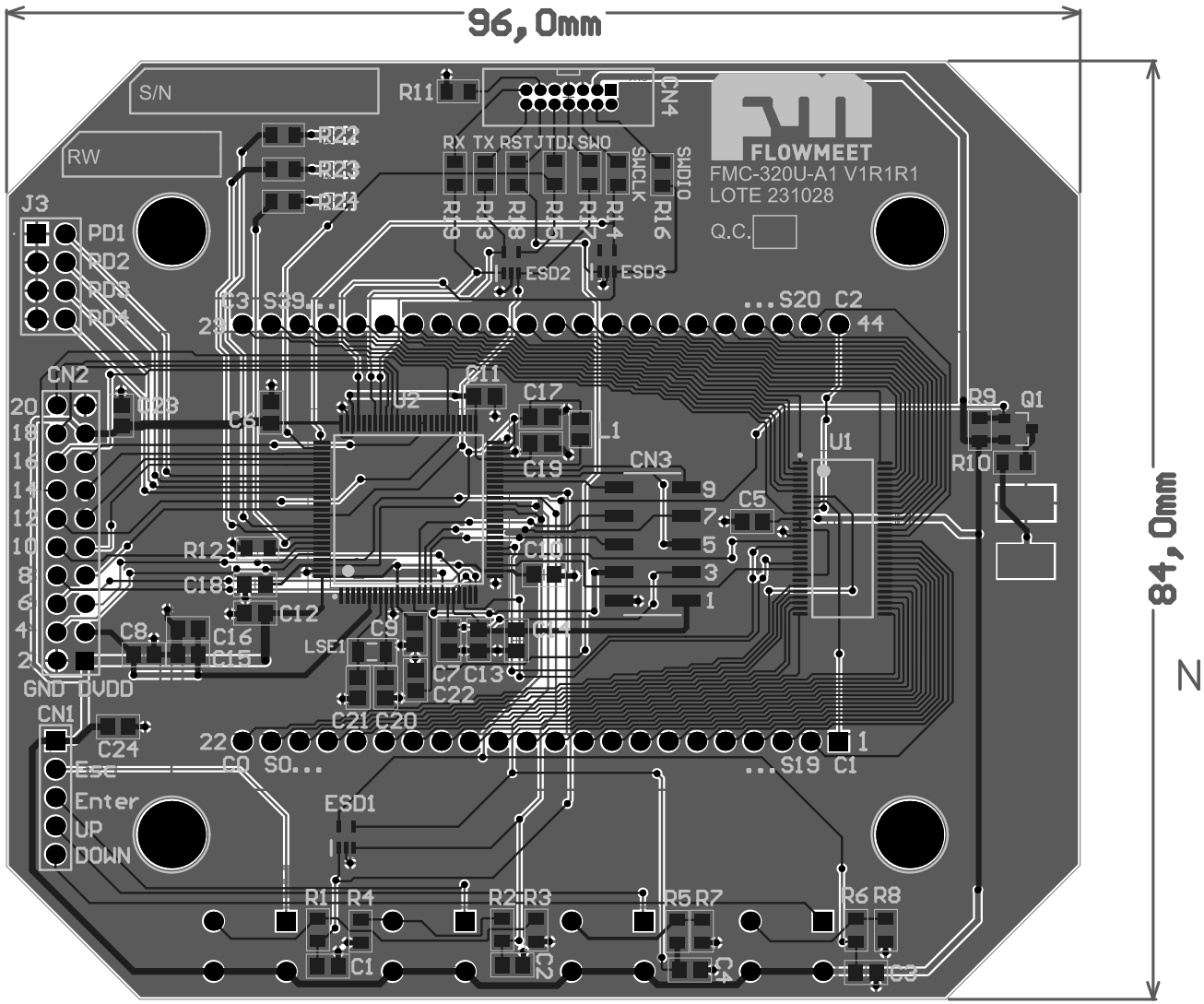
Puertos del uC conectados a un port undefined, tienen funcion sugerida pero no definida.



Setup de condiciones en power-on/Reset/Online




DVDD
CN4
1 2
3 4
5 6
7 8
9 10
11 12
13 14
T_SWDIO
T_SWCLK
T_SWO
T_JTDI
T_NRST
T_VCP_TX
R11
100R
T_VCP_RX
GND
GND
U2C
SWDIO
SWCLK
JTDI
72
76
77
PA13/JTMS/SWDIO
PA14/JTCK/SWCLK
PA15/JTDI
NRST
14
NRST
C22
100nF
GND
PB13/JTDO/SWO
PB4/NJTRST
PH3-BOOT0
89
90
94
SWO
R12
10K
GND
T_VCP_TX
R13
47R
USART1_TX
USART1_TX
T_JTDI
R15
47R
JTDI
T_SWO
R17
47R
SWO
T_VCP_RX
R19
47R
USART1_RX
USART1_RX
1 3 4 5
ESD2
2
GND
T_SWCLK
R14
47R
SWCLK
T_SWDIO
R16
47R
SWDIO
T_NRST
R18
47R
NRST
1 3 4 5
ESD3
2
GND



FLOWMEET S.R.L.		
Board ID FMC-320U-A1	Software Altium Designer v20.2.4	
Date 31/10/2023	File Name FMC-320U-A1.PcbDoc	Engineer DHS

Notes:

1) SMD diodes/LEDs mark is for CATHODE.
2) All capacitors are ceramic with no polarity.